Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 44914

Bachelor of Computer Application (Third Semester) (Main/Back) Examination, 2025-26

BCA-203N

Numerical Analysis and Statistical Techniques

Time : Two Hours] *[Maximum Marks : 75*

Note : Attempt **all** the sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt all **ten** questions. Each very short question carries **2** marks. 2×10=20

1. (a) What is numerical analysis?

   (b) Define error and types of errors.

   (c) What is an iterative method? Give an example.

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(d) Define linear regression.

(e) Write the formula for the bisection method.

(f) What is Newton's forward interpolation formula?

(g) Define continuous random variable.

(h) What is random variable?

(i) What is probability?

(j) Define Trapezoidal Rule.

Section-B

(Short Answer Type Questions)

Note : Attempt any **two** questions. Each short question carries **15** marks. Short answers are required not exceeding **200** words.

15×2=30

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2. Explain the Newton-Raphson method with proper example.

3. What is numerical differentiation? Explain how first and second derivatives can be obtained using forward difference formulas.

4. Given:

x:	1	2	3	4
y:	1	8	27	64

Find y at x=2.5 using Newton's Forward Interpolation problem.

5. Derive Simpson's $\frac{1}{3}^{rd}$ rule. Also compare it with Trapezoidal rule.

Section-C

(Long Answer Type Questions)

Note : Attempt any **one** question. Each long question carries **25** marks. Long answers are required not exceeding **800** words.

$25 \times 1 = 25$

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6. Explain Ruge-Kutta method for 1st and 2nd order differential equations.

7. Find the roots of $f(x) = x^3 - 4x - 9$ using Bisection method.

8. Evaluate $\int_0^2 (x^2 + 1)dx$ using Trapezoidal Rule.

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